

A grayscale photograph of an industrial manufacturing environment. Several large, white robotic arms are positioned over a car chassis. The background shows a complex network of steel beams and industrial lighting. In the foreground, a person's hand is holding a tablet that displays a digital interface with green and blue 3D models of robotic arms and a line graph.

Private LTE/5G Networks: A Catalyst for Industry 4.0

Abstract:

This paper covers how cellular technology is now expanding its footprint in the private network space. Cellular technology is enabling the unification of IT and OT domains in factories thus redefining how factories and enterprises are managed in the Industry 4.0 era. The adoption of cellular technology is being precipitated by new technologies, spectrum availability and industry bodies focusing on standardizing interfaces.

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Table of Content

Introduction - The Need for Private Cellular Networks	04	LTE or 5G	08
Use Cases for Private Cellular Network	05	Wi-Fi 6	08
Factors Accelerating Cellular Adoption for Enterprises	06	Private Network Deployment Models	09
a. Reducing Equipment Cost and Size		Examples of Deployments	10
b. Cost of Deployment		Summary	11
c. Ease of Deployment and Management		Abbreviations	12
d. Unlicensed Spectrum		References	13
e. Dedicated Enterprise Spectrum		About the Author	14
f. Small Cells with SON Capability		About Sasken	14
g. Network Slicing			
h. Open RAN (O-RAN)			
i. Multi-access Edge Computing (MEC)			



Introduction - The Need for Private Cellular Networks

The private LTE and 5G market is slated to reach \$4.45 billion by 2025 with Industrial IoT and enterprise networks contributing a major portion of this revenue [1]. Private network is not a new phenomenon as it has always existed. For example, critical communications networks are private networks. Similarly, enterprises have had their private networks, be it wired or wireless. What is new however is the adoption of cellular technology (4G/ 5G) to build these private networks. This inclination of the industry towards cellular technology has been caused by multiple reasons. From the need to avoid getting locked into a particular vendor or technology, to ensuring easy maintenance and upgrading, and also wanting to lower costs. Cellular technology fits these requirements very well. Cellular technology has been in existence for nearly three decades now, with an active standards body, a wide ecosystem with plenty of choices and the ability to custom build a network at a relatively low cost. Of course, the needs of industries have also evolved but legacy technologies have not been able to address the needs or match their pace in growth.

There is a growing need to connect a massive number of low-power devices to the internet. This requires a technology which can connect a huge number of devices in a scalable, dynamic, and cost-effective manner. [2] In factories, there is a need for unified communications, i.e. handling voice and data communications on a common infrastructure. Legacy implementation is a cocktail of wired and wireless protocols such as WirelessHart, ZigBee, Ethernet, Modbus, etc. [3] This is where cellular steps in. Along with voice communications, LTE/5G also addresses the data communication aspects. LTE networks can provide coverage of 60,680 LPWA devices per square kilometer. 5G networks must meet the ITU standard of 1 million devices per square kilometer. [2]

With an increasing number of devices and people connected to the network, there is also a need to protect data. Private cellular network owners or operators can control the parties who have access to this data. These companies can therefore ensure no secure information leaves the premises. [4]

In the industrial setting such as mines, warehouses, and shipyards the range of connectivity can be an issue. Wi-Fi cannot scale as it needs too many access points and associated cabling. For example, it may not be possible to run cables for Wi-Fi at mining sites due to safety issues. Having a wired connection can also be very expensive, not to mention that it can be very difficult to maintain as well. Cellular technology on the other hand, has a wider coverage range and can be deployed at lower costs, compared to Wi-Fi and other wired connections.



Use Cases for Private Cellular Network

Transportation Hubs/Warehouses/Utilities

- Video surveillance is possible due to high bandwidth availability
- Easier to gather telemetry data over long ranges

Oil Fields/Mines

- Mines are in remote areas with no public cellular network
- Cabling is expensive and the problem is compounded by a hazardous environment
- For wide areas, it is unsuitable for Wi-Fi or any short-range communication
- Ensures critical communications for safety and operations team enabled by low latency and reliability of cellular networks compared to Wi-Fi

Manufacturing

- Provides unified communication services for both voice & data
- Cabled networks are costly, especially if it spans several kilometres

- Ensures timely critical communication for both safety and operations team

Retail

- Can be utilized for PoS terminals, security monitoring, and for digital signage screens

Public Safety

- 5G networks can provide the low latency and high reliability needed for critical communication
- LTE or 4G offers the option of transmitting high bandwidth data such as videos
- 4G/5G bandwidths also offer high precision location services



Factors Accelerating Cellular Adoption for Enterprises

a) Reducing Equipment Cost and Size

Virtualization is democratizing the Radio Access Network (RAN) market by lowering the entry barriers to Network Equipment Makers (NEMs). Virtualization enables the use of COTS hardware which lowers the cap-ex and op-ex and equipment footprint, provides deployment flexibility, and makes it easier to configure or upgrade. [6]

b) Cost of Deployment

Private LTE network deployment costs lesser than Wi-Fi deployment. There are companies which are claiming to deploy mission critical private LTE at 50% of the cost of deploying an enterprise Wi-Fi system. Also, the number of access points required for LTE is 25% of the number needed for a Wi-Fi connection. [3]

c) Ease of Deployment and Management

Operators, NEMs, and system integrators will provide network services as a managed service, which will reduce the friction for enterprises towards deploying the networks. Operators may directly offer the services when there is a need of spectrum resources and the deployment is large-scale. In such cases, the NEMs and/or system integrators will coordinate with the operator to deploy the network in a licensed spectrum. However, NEMs or system integrators may provide managed services when network deployments are small-scale or purely working on an unlicensed spectrum. Once deployed, the ease in its management is also crucial. If the enterprise wants to change something, there should be easy configurability which can be done by the local IT team.

d) Unlicensed Spectrum

CBRS and 5 GHz spectrum can bring down operational costs (as operators don't have to pay for spectrum) and free up spectrum usage for private LTE or 5G purposes. 5G NR systems are in development for several shared bands. A decision was made to include support for 5G NR unlicensed spectrum in 3GPP Release 16 during the 3GPP plenary meeting. The NR-U work item that was just approved by 3GPP supports both existing 5GHz unlicensed spectrum band and the new "greenfield" 6GHz unlicensed spectrum band. [7]

e) Dedicated Enterprise Spectrum

In some markets, regulators are investigating or already allocating licensed spectrum to business verticals for running private networks, particularly for industrial IoT. This typically involves licenses with small geographic areas to ensure reuse and access to spectrum for necessary companies. It gives the industrial houses surety towards tenure and predictability of network performance. This makes it ideal for Industrial IoT purposes. [8]

f) Small Cells with SON Capability

Small cells with the Self-Organizing Network (SON) feature make it very easy to deploy a network. Self-organizing network automation technology makes the planning, configuration, management, optimization, and healing of mobile radio access networks simpler and faster, leading to lower costs. [9] Sprint's "Magic Box" manufactured by



Airspan Networks was easily installed by the customer in a matter of minutes, and automatically connected to a nearby Sprint cell site. [10]

Large enterprises and campuses can be covered with fewer small cells. Several commercially available indoor small cells easily cover over 10,000 ft. CBRS spectrum rules allow outdoor small cells to transmit at as high as 50W, which is sufficient to cover few sq.kms. [11]

g) Network Slicing

Network slicing is the creation of logical networks rendering different services using shared infrastructure elements. Slicing helps support new use cases and differentiated experiences, including private networks and will be a source of new revenue for the mobile operators. [12] For example, by using the same network infrastructure an operator can provide multiple services such as IoT, eMBB or URLLC.

h) Open RAN (O-RAN)

The O-RAN alliance was started by a consortium of cellular operators with the objective of developing open RAN architecture, virtual network elements and espousing open interfaces. Doing so would increase equipment interoperability and reduce costs by enabling a vibrant supplier ecosystem. [13]

Groups like Telecom Infra Project (TIP) led by Facebook are also working on open, interoperable and agile systems in order to make connectivity ubiquitous.

i) Multi-access Edge Computing (MEC)

Multi-access Edge Computing (MEC) transforms mobile networks into distributed cloud computing platforms that operate at the access network. MECs can be used to solve three basic problems [1][14]:

Latency: Through MEC, a subscriber's experience with an application can be enriched with low latency and agile applications. MEC is specifically designed to reduce transport latency by deploying computing and storage capabilities closer to where content and services are created or consumed.

Security: By using local breakouts, business-critical applications can run in a secure manner without requiring traffic to be routed through a centralized mobile core

Network costs: It is not economically viable to serve high volume content from a centralized location. Operators can access their Radio Access Network (RAN) edge to authorized third parties. This allows flexibility and rapidness in deploying innovative applications and services towards mobile subscribers, enterprises and vertical segments. Use cases include: video analytics, location services, Internet-of-Things (IoT), augmented reality, optimized local content distribution and data caching. [15]



LTE or 5G

Choosing which technology to deploy between LTE and 5G depends on the use case. According to Nokia, the LTE-versus-5G debate is less about reliability, and more about network latency and its flipside-measure, network bandwidth. For most use cases LTE may be sufficient to utilize. Privately managed LTE provides latency of 40-50 milliseconds and with fine tuning may even reach 10 milliseconds. 5G with tuning will eventually go further by going closer to 1 millisecond.

Consultancy firm KPMG reckons cellular at large has failed in factories. Ethernet is the go-to technology for critical communications, with Wi-Fi being a compromise for moving parts, and cellular remaining as a public network on the fringes. Cellular has also been tested, and found wanting as a technology by KPMG. [16]

Wi-Fi 6

Cellular networks still need to compete with Wi-Fi which has gotten better over the years. Wi-Fi 6 (IEEE 802.11ax) will have a single-user data rate that is 37% faster than 802.11ac. It will also offer four times the throughput per user in crowded environments along with better power efficiency which should translate to a boost in device battery life. [17]

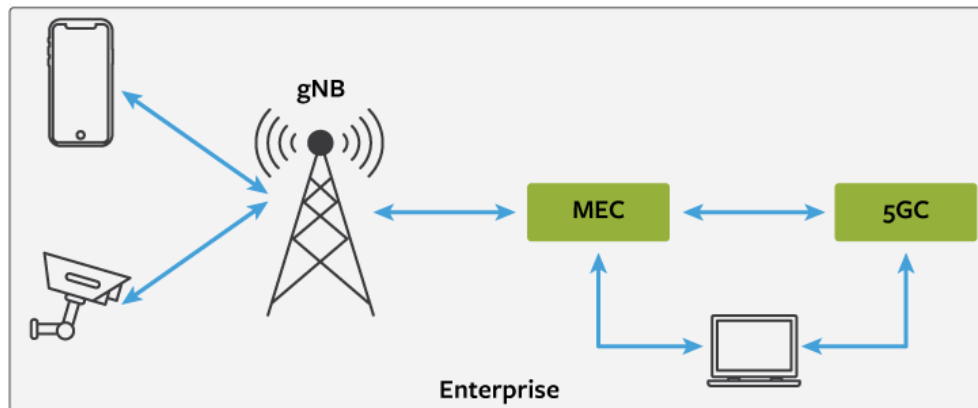
5G and Wi-Fi 6 represent different approaches to wireless connectivity. Yet, both are based on several of the same technological building blocks (e.g., OFDM, MIMO, and higher-order modulation). Like all cellular technologies, 5G is appropriate when the user requires macro coverage with mobility and can afford the additional costs for these capabilities. Wi-Fi 6 has excellent in-building mobility but does not roam well at high speeds. More importantly, Wi-Fi 6 is at least on par with 5G, in terms of throughput, latency, spectral efficiency, and connection density. [18]



Private Network Deployment Models

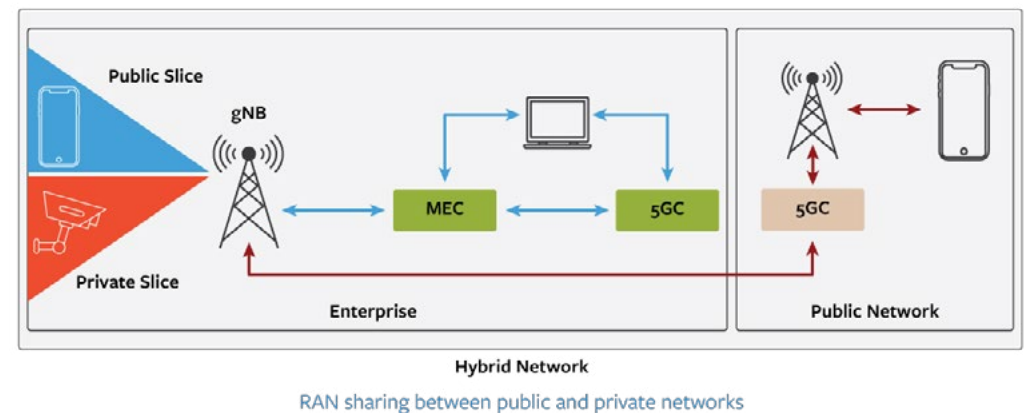
Public cellular networks are deployed and operated by a licensed cellular operator. The operator owns the RAN and the core network. However, in the case of private networks the deploying, operating, and owning of the network may be handled by different entities. Broadly there are two basic forms of private networks. [8]

- Independent private networks without dependencies on a licensed public operator or a wide-area network. Integration with the public network is possible as an optional choice.



Independent Private Network

- Private networks deployed in conjunction with a public network, allowing various levels of integration. Dedicated Core Networks (DÉCOR) employs virtualization and slicing mechanisms to carve out a portion of the network for private use. This allows employees of enterprises to access their data securely even when they exit the enterprise premises. [4][8]



Examples of Deployments

Nokia has deployed private LTE with 120 customers spread across multiple industries. [19] Nokia and China Unicom have also deployed a private LTE network for smart manufacturing services at a BMW plant in China. The Nokia Virtualized Multi-access Edge Computing (vMEC) solution will leverage China Unicom's 4G LTE network to provide low-latency support for smart manufacturing activities at the plant. [20]

Rio Tinto, the mining conglomerate was one of the first large enterprises to use a private LTE network to support commercial operations at scale. In its initial deployment, Rio Tinto was able to replace 30 Wi-Fi access points with just four LTE base stations outside of the blast zone. LTE allowed them to prioritize safety and production critical traffic over corporate traffic. [4]

Air France also implemented a private LTE at Charles-de Gaulle airport in Paris with Ericsson installing, commissioning and integrating the network. [1]



Summary

Enterprises working in a multiple industries need wireless connectivity that offers high bandwidth and predictable latency, with cost-efficient support for higher density of client devices. It should also be secure and reliable in enabling critical business operations while being fully controlled by the enterprise. Private LTE and 5G networks can help meet these requirements.

Private LTE networks utilize the same standards, network architecture, and supplier ecosystem used by mobile operators presently. Recent advancements make it possible for enterprises to own and operate private LTE networks. This includes the development of enterprise small cells for deploying over existing Ethernet LAN, software to automatically configure small cells, virtualized core network products running as software on off-the-shelf hardware, and availability of unlicensed spectrum. [11] Private cellular networks also provide operators a new revenue source.

The arrival of cellular into enterprises also does not necessarily mark the end of Wi-Fi, as LTE and 5G can co-exist with Wi-Fi for some time. Wi-Fi and cellular (5G) are both evolving to better serve end users, and both markets will grow to serve the macro trend of connecting and analyzing devices. Wi-Fi will continue to prove its value as a reliable, secure, and cost-effective wireless access technology for most enterprise applications, as it currently does. 5G will serve applications requiring mobility and macro range, and for certain industrial use cases where customers would prefer physical network segmentation. [18]



Abbreviations

3GPP	Third Generation Partnership Project	MNO	Mobile Network Operator
CAP-EX	Capital Expenditures	NEM	Network Equipment Manufacturer
CBRS	Citizens Broadband Radio Service	NFV	Network Function Virtualization
COTS	Commercial off-the-shelf	NR	New Radio
C-RAN	Centralized Radio Access Network	NR-U	New Radio- Unlicensed Spectrum
eMBB	Enhanced Mobile Broadband	OFDM	Orthogonal frequency-division multiple
eNB	Evolved NodeB	Op-ex	Operating Expense
IoT	Internet of Things	PoS	Point of Sale
ITU	International Telecommunication Union	RAN	Radio Access Network
LPWA	Low Power Wide Area	SON	Self Organizing Networks
LTE	Long Term Evolution	SD-WAN	Software Defined Wide Area Networks
MEC	Multi-Access Edge Computing	URLLC	Ultra Reliable and Low Latency Communications
MIMO	Multiple-input, multiple-output	vRAN	Virtual Radio Access Network



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About Author

Pradeep has a decade of experience in the electronics and communications industry. At Sasken, he works for the Product Engineering Practice and is responsible for developing and marketing new services for the cellular technology industry.

About Sasken

Sasken is a specialist in Product Engineering and Digital Transformation providing concept-to-market, chip-to-cognition R&D services to global leaders in Semiconductor, Automotive, Industrials, Smart Devices & Wearables, Enterprise Grade Devices, SatCom, and Transportation industries. For over 30 years and with multiple patents, Sasken has transformed the businesses of over a 100 Fortune 500 companies, powering over a billion devices through its services and IP.



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